

The Program in Action

When you run *factorfinder.py*, the output will look like this:

```
Factor Finder, by Al Sweigart al@inventwithpython.com
--snip--
Enter a number to factor (or "QUIT" to quit):
> 26
1, 2, 13, 26
Enter a number to factor (or "QUIT" to quit):
> 4352784
1, 2, 3, 4, 6, 8, 12, 16, 24, 29, 48, 53, 58, 59, 87, 106, 116, 118, 159,
174, 177, 212, 232, 236, 318, 348, 354, 424, 464, 472, 636, 696, 708, 848,
944, 1272, 1392, 1416, 1537, 1711, 2544, 2832, 3074, 3127, 3422, 4611, 5133,
6148, 6254, 6844, 9222, 9381, 10266, 12296, 12508, 13688, 18444, 18762, 20532,
24592, 25016, 27376, 36888, 37524, 41064, 50032, 73776, 75048, 82128, 90683,
150096, 181366, 272049, 362732, 544098, 725464, 1088196, 1450928, 2176392,
4352784
Enter a number to factor (or "QUIT" to quit):
> 9787
1, 9787
Enter a number to factor (or "QUIT" to quit):
> quit
```

How It Works

We can tell if a number is a factor of another number by checking if the second number evenly divides the first number. For example, 7 is a factor of 21 because $21 \div 7$ is 3. This also gives us another of 21's factors: 3. However, 8 is not a factor of 21 because $21 \div 8 = 2.625$. The fractional remainder component tells us this equation did not divide evenly.

The % mod operator will perform division and tell us if there's a remainder: $21 \% 7$ evaluates to 0, meaning there is no remainder and 7 is a factor of 21, whereas $21 \% 8$ evaluates to 1, a nonzero value, meaning that it isn't a factor. The factor finder program uses this technique on line 35 to determine which numbers are factors.

The `math.sqrt()` function returns the square root of the number passed to it. For example, `math.sqrt(25)` returns 5.0 because 5 times itself is 25, making it the square root of 25.

```
1. """Factor Finder, by Al Sweigart al@inventwithpython.com
2. Finds all the factors of a number.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, beginner, math"""
5.
6. import math, sys
7.
8. print('Factor Finder, by Al Sweigart al@inventwithpython.com
9.
10. A number's factors are two numbers that, when multiplied with each
```